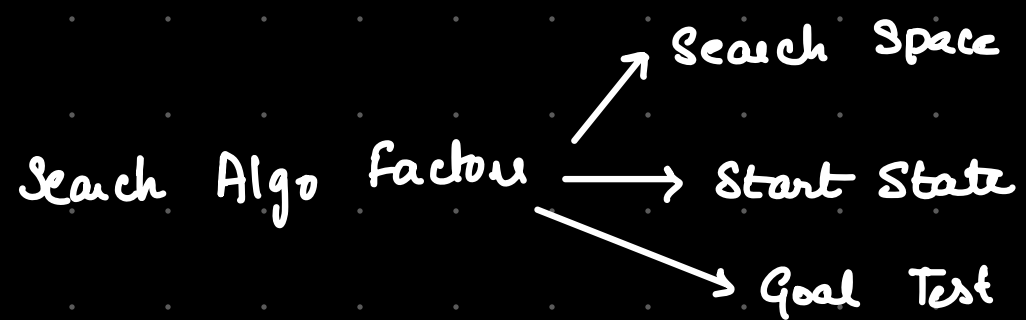




Problem Formulation:

- ① Initial State
- ② Actions
- ③ Transition Model
- ④ Goal Test
- ⑤ Path cost funcn.

..

Water Jug Problem

$x \rightarrow 4$ gallon jug

$y \rightarrow 3$ gallon jug

Goal = 2 gallon out of the 4 gallon jug

$x = 4L$

$y = 3L$

$x = 0 \quad y = 0$

$x = 0 \quad y = 3$

$x = 3 \quad y = 0$

$x = 3 \quad y = 3$

$x = 4 \quad y = 2$

$x = 0 \quad y = 2$

$x = 2 \quad y = 0$

$y < 3$

$x < 4$

$y < 3$

$x < 4$

$x = 4$

$x < 4$

Goal state

Fill y to brim

Full content of

$y \rightarrow x$

Fill y to brim

$y \rightarrow x$

Empty x

$y \rightarrow x$

Uninformed Search



- no prior knowledge abt states than problem defn.
- generate succ. & distinguish b/w goal & non-goal

1. BFS

2. DFS

3. DLS (Depth Limited)

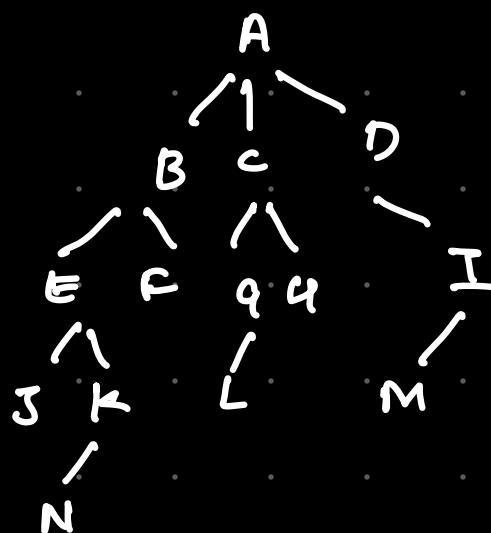
4. IDDS (Iterative Deepening Depth FS)

BFS

Adv	D. Adv
soln. given if exists	a lot of mem. each level $\Rightarrow$ expand next
minimal if mult. soln.	lot of time if far from root node

BFS = Queue

Q.

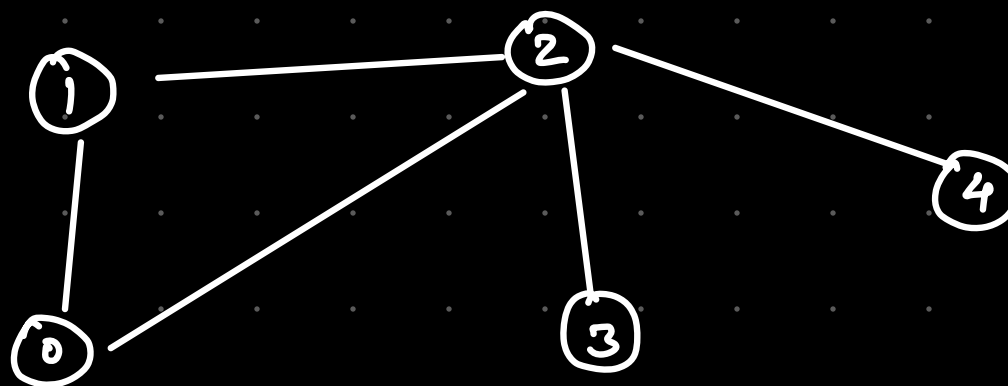


TC: $O(V+E)$

Queue : ~~A~~ ~~B~~ ~~C~~ ~~D~~ ~~E~~ ~~F~~ ~~G~~ ~~H~~ ~~I~~ ~~J~~ ~~K~~ ~~L~~ ~~M~~ ~~N~~

A, B, C, D, E, F, G, H, I, J, K, L, M, N

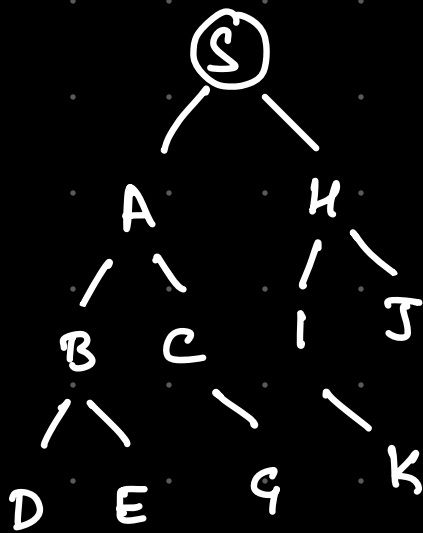
Q.



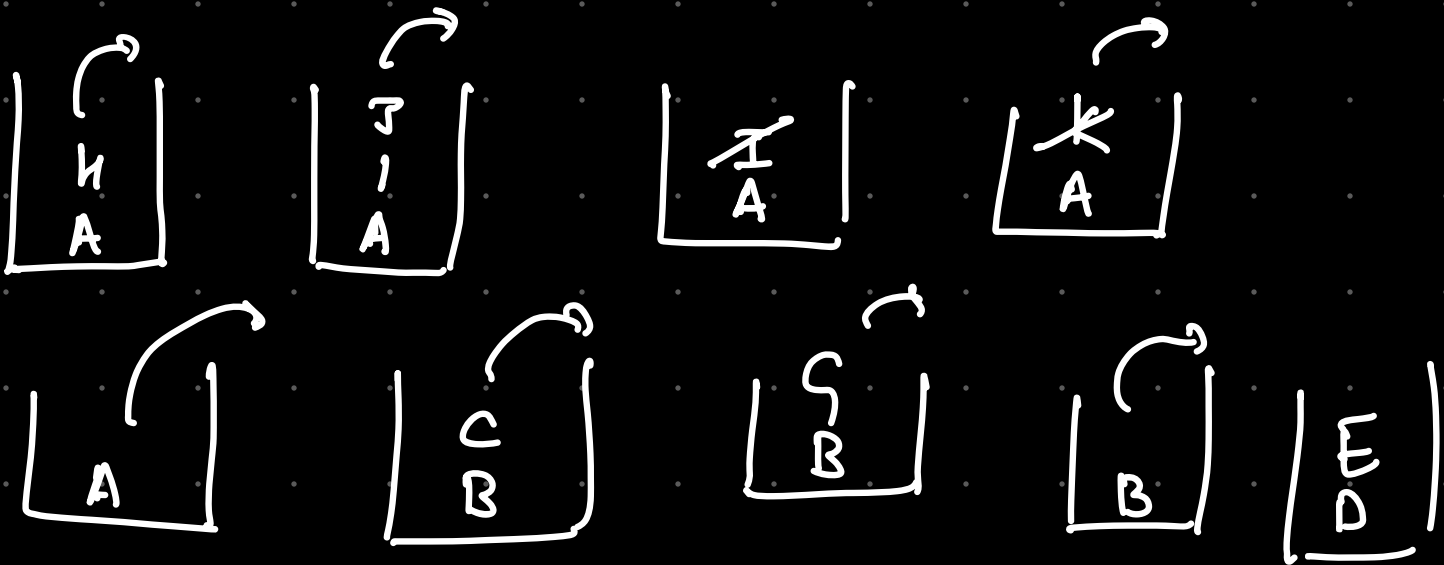
Queue: $\emptyset$ / 2 / 3 / 4

0, 1, 2, 3, 4

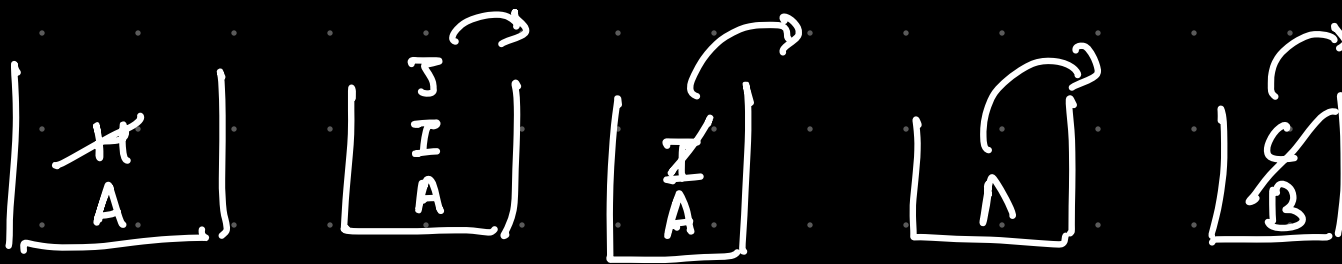
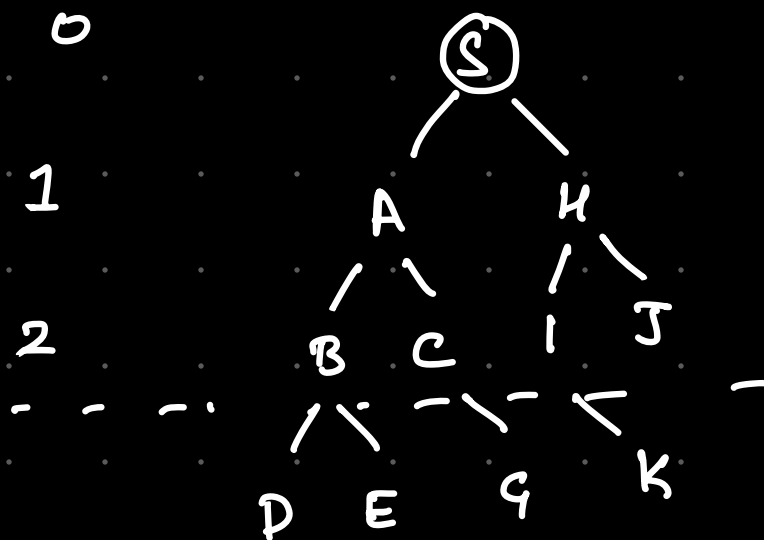
2. DP S



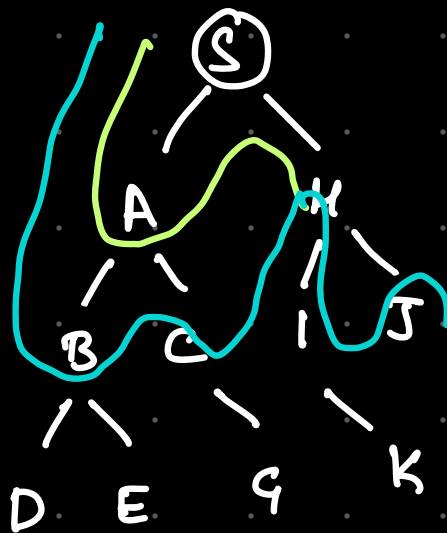
$S \rightarrow H \rightarrow I \rightarrow I \rightarrow$
 $K \rightarrow A \rightarrow$
 $C \rightarrow G \rightarrow$
 $B \rightarrow E$
 $\rightarrow D$



3. DLS



$S \rightarrow H \rightarrow I \rightarrow A \rightarrow C \rightarrow B$

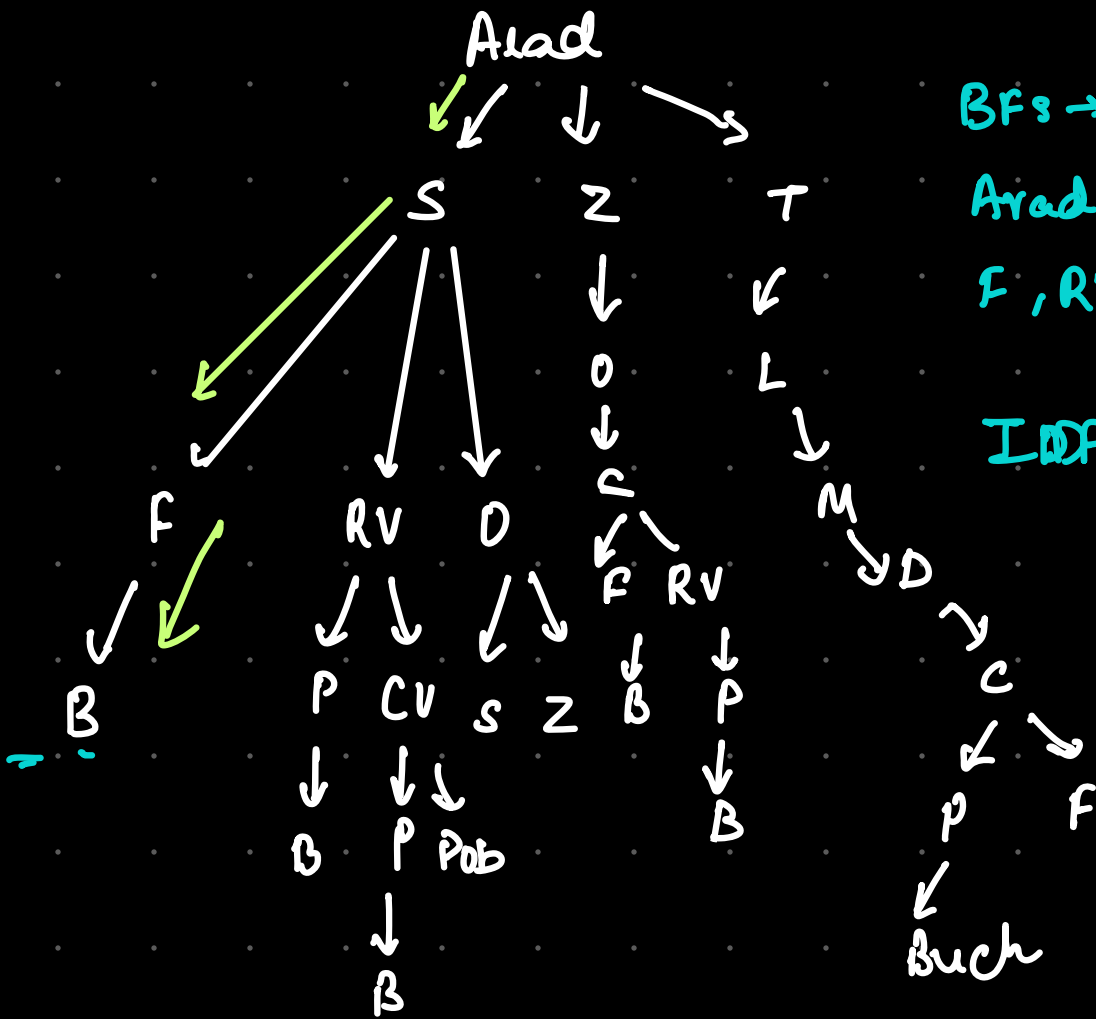


Level 0 = S

Level 1 = S → A → H

Level 2 = S → A → B → C → H → I → J

Map of Romania



BFS →

Arad, Sib, Zerind, Tim
F, RV, O, L, F, B

IOFS $\rightarrow A, S, Z, T,$
F, RV

Informed Search

$f(n)$ = total cost estimate

Best First Search

expand n' with lowest $f(n)$



$S \rightarrow B \rightarrow F \rightarrow G$

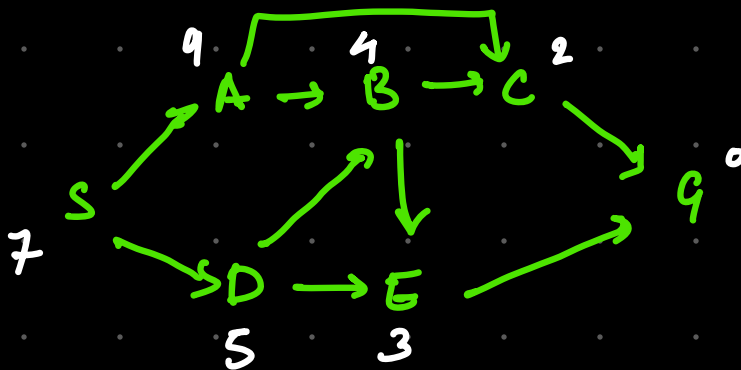
$f(n) = 1 + 3 + 1 = 5$

Q BFS

$$f(n) = h(n)$$

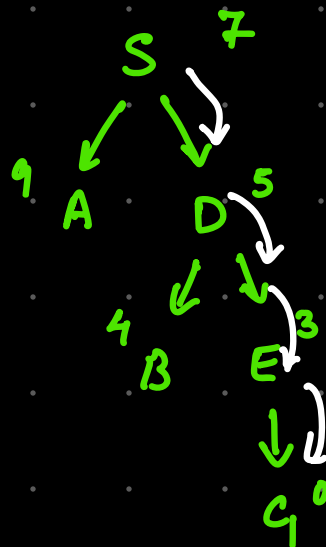
↳ expands node that appears to be closer to goal.

Q.



$$f(n) = h(n)$$

$S \rightarrow D \rightarrow E \rightarrow G$



Straight line as a heuristic = admissible

A* Search

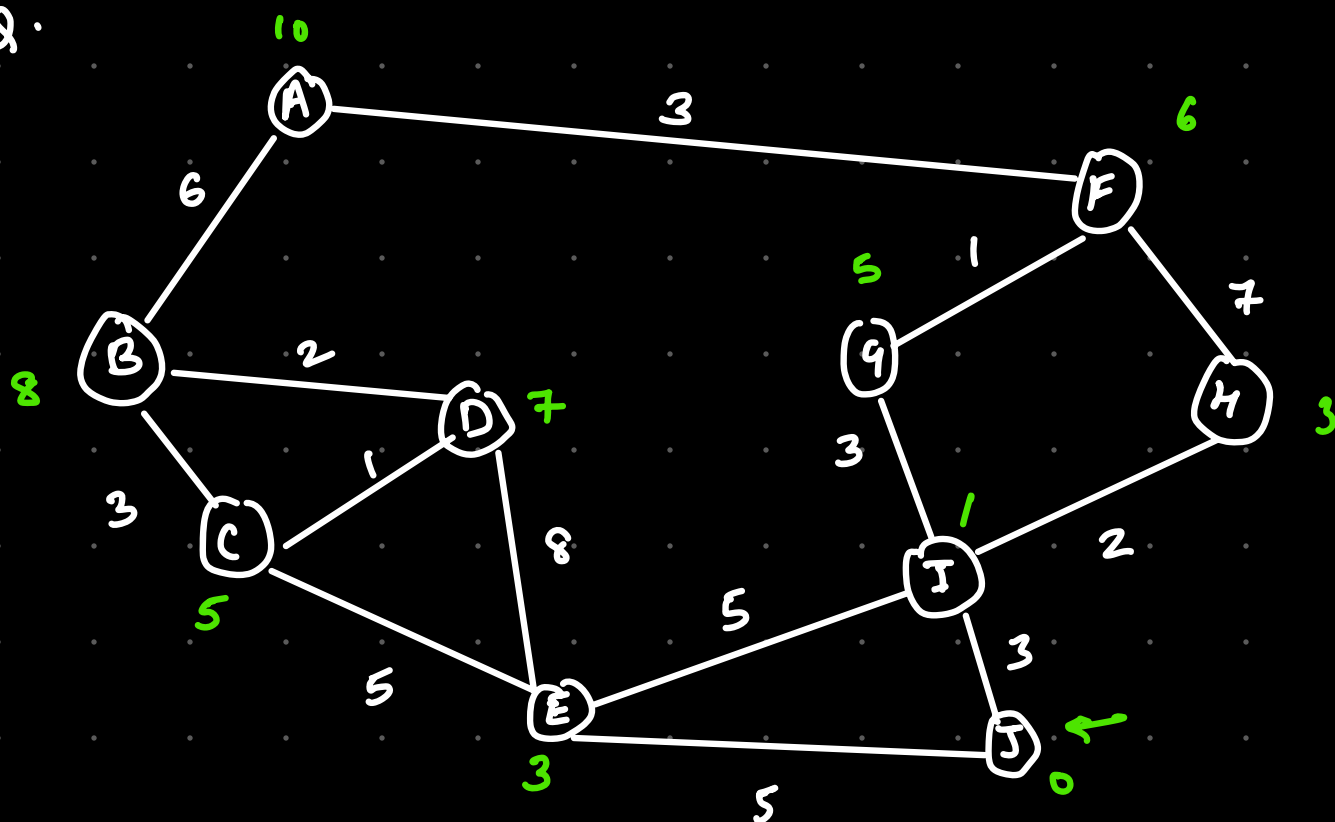
$$f(n) = g(n) + h(n)$$



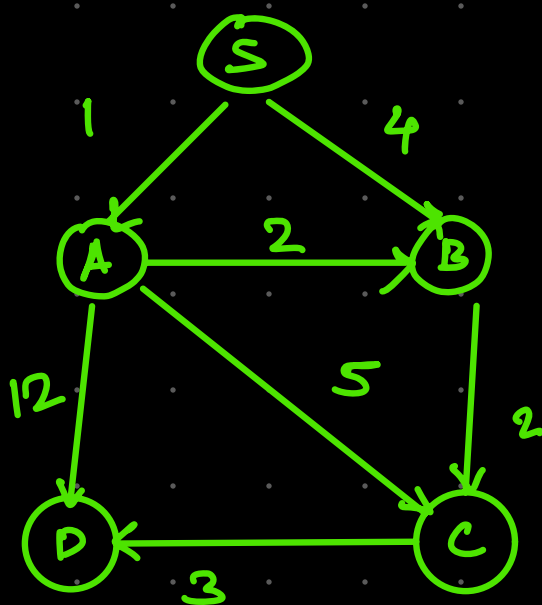
cost to
reach n from
start

↪ heuristic

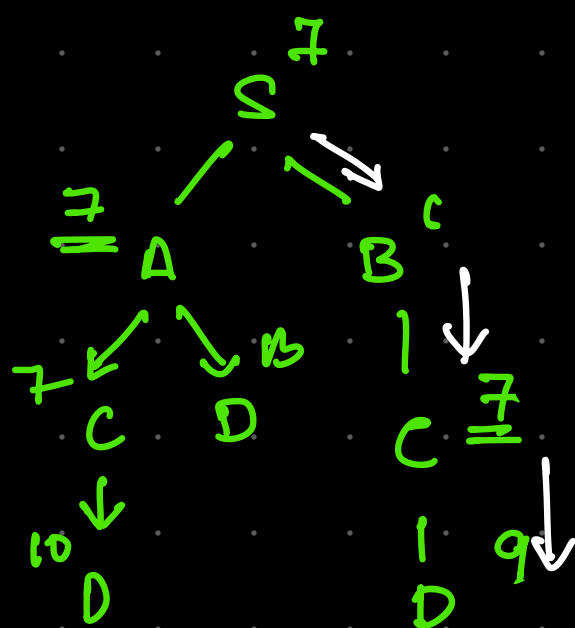
Q.



Q.



S	7
A	6
B	2
C	1
D	0



$S \rightarrow B + C \rightarrow D$

cost = 9

8-puzzle

Heuristic:

$$f(n) = h_1(n) + h_2(n)$$

↑
no. of
misplaced
tiles

Manhattan Dist. =
sum of dist of
every misplaced tile from
its posn.

N-Queens

$h(n)$ = no. of
attacking
Q. p

Graph Col.

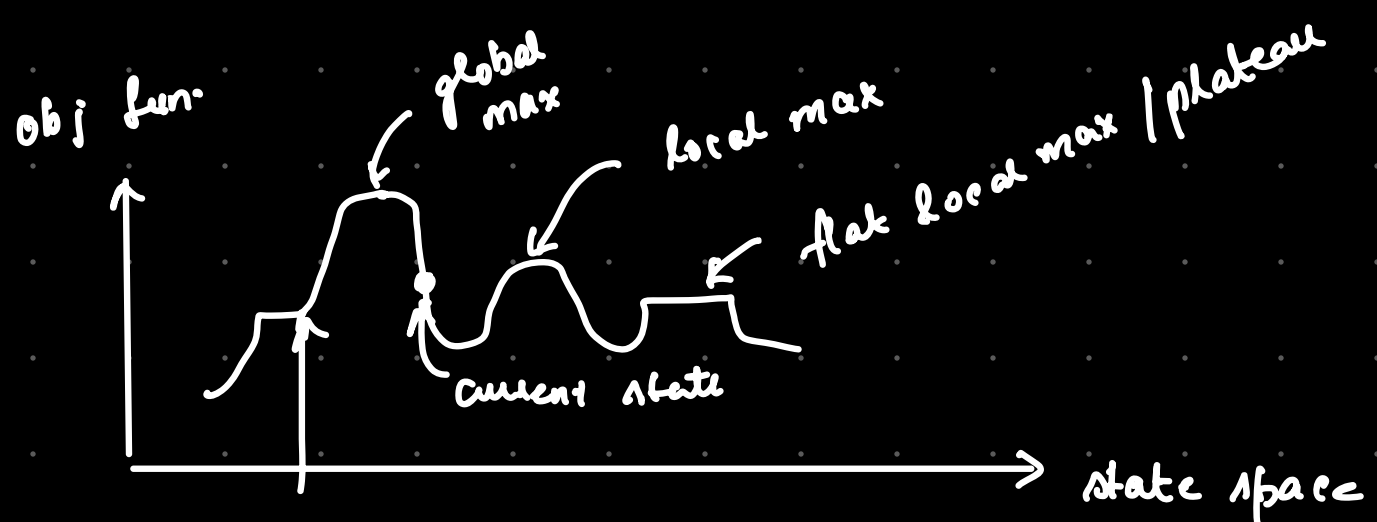
no. of adj node
pairs that have
same col.

Path doesn't matter = LOCAL search

↓
optimize search space

→ Heuristic method of solving optimization problems

→ moves soln → soln in search space of candidate soln. by applying local changes, until a soln. deemed optimal is found / time elapsed



Local maxima

↳ a state better than its neighbouring states but there exists a state better than it.

Global maxima: best possible state in the state space

↙
obj. func. = highest value.

Plateau / flat local maxima: a region where neigh. states have the same value

↘
obj. func. =

Ridge: a region higher than neigh. but itself has a slope

↳ special kind of local max.

Shoulder: is a plateau with an uphill edge.

Hill Climbing Problem



→ is a local search algorithm which is just loop that continually moves in the upward dirn / increasing val.

→ it terminates when it rea

CSP $\rightarrow$ Constraint Satisfaction Problem

Goal = reach a state that satisfies some set of constraints

States defined by V_i
 $\hookrightarrow$ Domain D_i

4-Queens

$$V_i = \{Q_1, Q_2, Q_3, Q_4\}$$

$$D_i = \{1, 2, 3, 4\}$$

Constraint

$$Q_i \neq Q_j$$

$$|Q_i - Q_j| \neq |i - j|$$

Crypt Arithmetic

- carry can only be 1
- assign numeric value to char.
- no two letters have the same value.

Eg:
$$\begin{array}{r} \text{TWO} \\ + \text{TWO} \\ \hline \text{FOUR} \end{array}$$

Column Eqn

$$2O = R + 10C_1$$

$$2W + C_1 = U + 10C_2$$

$$2T + C_2 = O + 10C_3$$

$$\boxed{C_3 = 1 \quad f = 1}$$

TWO
TWO
FOUR

$$f = 1$$

	$20 = R + 10C_1$		
0	$R = 0$		X
1	$R = 2$	$C_1 = 0$	X
2	$R = 4$	$C_2 = 0$	
3	$R = 6$		
4	$R = 8$		
5	$R = 0$	$C_1 = 1$	
6	$R = 2$	$C_1 = 1$	
7	$R = 4$	$C_1 = 1$	
8	$R = 6$	$C_1 = 1$	
9	$R = 8$	$C_1 = 1$	

Assume $0 = 5$

$$R=0 \quad C_1=1$$

$$\Rightarrow 2W + C_1 = u + 10C_2$$

$$2W + 1 = u + 10C_2$$

W			
0	$u=1$	$C_2=0$	X
2	$u=5$	$C_2=0$	X
3	$u=7$		
4	$u=9$		
6	$u=3$	$C_2=1$	
7	$u=5$	$C_2=1$	X
8	$u=7$	$C_2=1$	X
9	$u=9$	$C_2=1$	

$$\begin{array}{r}
 T w 0 \\
 Tw 0 \\
 \hline
 FOUR
 \end{array}
 \Rightarrow
 \begin{array}{r}
 \overset{1}{T} w 5 \\
 Tw 5 \\
 \hline
 15 \quad 0
 \end{array}$$

$\therefore w$ produces carry

Assume $w = c$

$$u = 3 \quad c_2 = 1$$

$$2T + c_2 = 0 + 10c_3$$

$\downarrow = 1$

$$2T + 1 = 5 + 10$$

$$2T + 1 = 15$$

$$\therefore T = 7$$

TWO
TWO
—
FOUR

1 1
7 6 5
7 6 5
—
1530

T 7
W 6
O 5
F 1
U 3
R 0

$$\begin{array}{r} \text{S E N D} \\ \text{M O R E} \\ \hline \text{M O N E Y} \end{array}$$

$$D + E = Y + 10C_1$$

$$N + R + C_1 = E + 10C_2$$

$$E + O + C_2 = N + 10C_3$$

$$S + M + C_3 = O + 10C_4$$

$$M = 1 \Rightarrow C_4 = 1$$

$$S + M + C_3 = O + 10$$

$$S + 1 + \underline{C_3} = O + 10$$

$\therefore 8, 9$

Assume 9,

$$\begin{array}{r} \text{S E N D} \\ \text{M O R E} \\ \hline \text{M O N E Y} \end{array}$$

$$\begin{array}{r} 9 \text{ E N D} \\ 10 \text{ R E} \\ \hline 10 \text{ N E Y} \end{array}$$

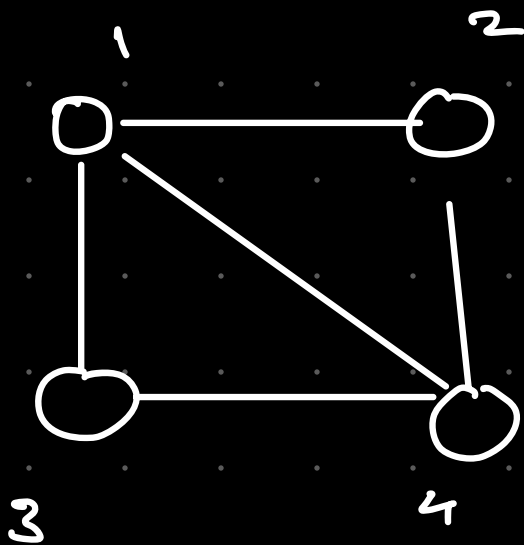
$$E + O + C_2 = N + \underline{\underline{10C_3}}$$

$$\underline{\underline{E + O + C_2}} = N \quad \swarrow_{\text{assumed} = 0}$$

2, 3, 4, 5, 6, 7, 8

↳

$$D + E = 7 + 10C_1$$

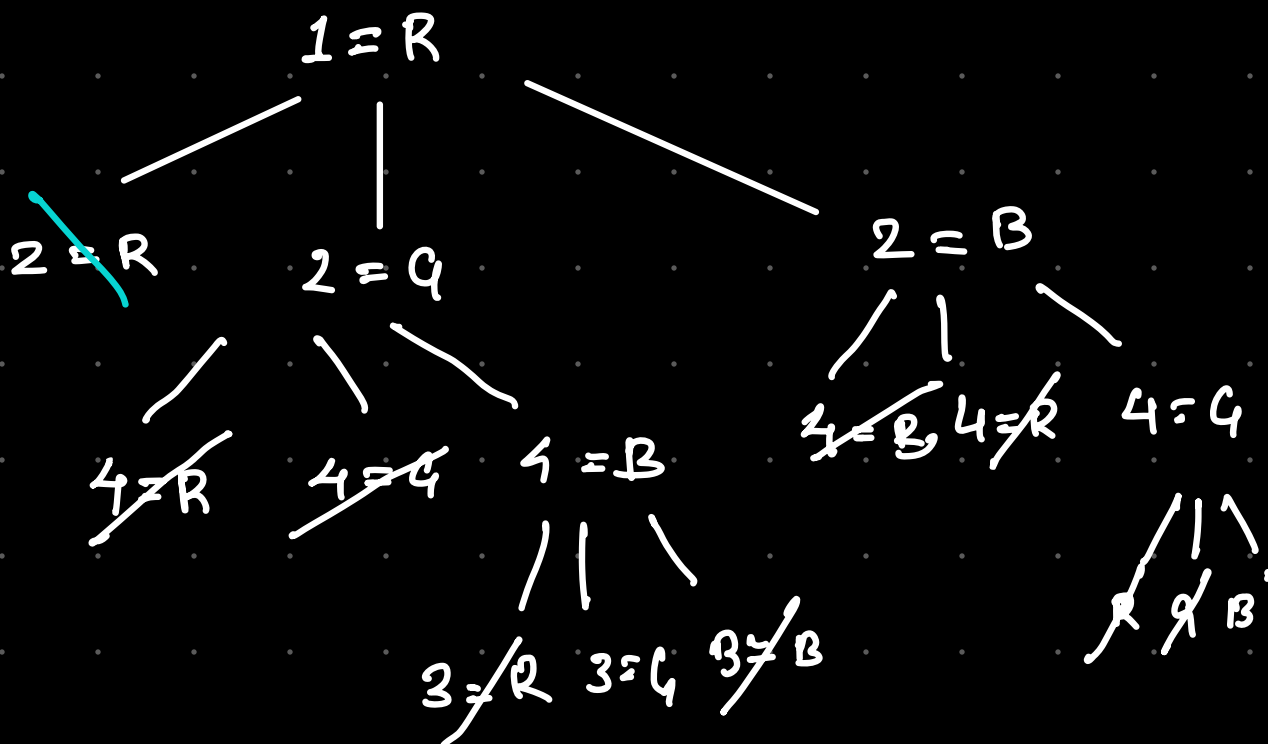


$$V = \{1, 2, 3, 4\}$$

$$D_i = \{R, G, B\}$$

$$C =$$

$$\{1 \neq 2, 1 \neq 3, 1 \neq 4, \\ 3 \neq 4, 2 \neq 4\}$$



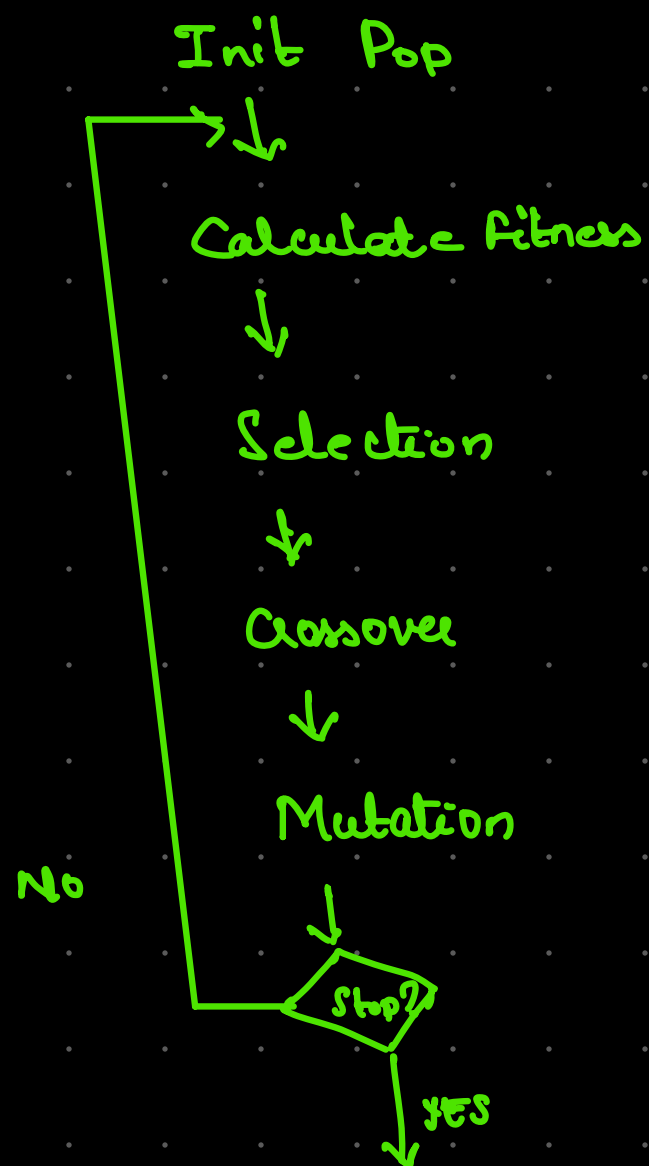
$$\therefore \text{If } \boxed{1 = R}$$

$$\begin{aligned} & \hookrightarrow 2 = G \\ & \quad 3 = G \\ & \quad 4 = B \end{aligned}$$

$$\text{or } \boxed{1 = R}$$

$$\begin{aligned} & 2 = B \\ & 3 = B \\ & 4 = G \end{aligned}$$

Genetic Algo



Q. 0/1 Knapsack

Item	1	2	3	4	5	6	7
p_i	6	5	8	9	6	7	3
w_i	2	3	6	7	5	9	4

$$W = 9$$

1. Chromosome Representation

Selected = 1 Not = 0

2. Initial Population

Generate random candidate soln.

	⁶ 2	⁵ 3	⁸ 6	⁹ 7	⁶ 5	⁷ 9	³ 4
C1	1	1	0	0	0	0	1
C2	1	0	1	1	0	0	0
C3	0	1	1	0	1	0	0
C4	1	0	0	1	0	0	0

3. Fitness Function

$$\text{Fitness} = \begin{cases} \sum p_i x_i \\ 0 \end{cases}, \sum w_i x_i \leq 9$$

	$\sum w_i x_i$	$\sum p_i x_i$	%
C1 =	9	14	0.482
C2 =	15	0	0
C3 =	14	0	0
C4 =	9	15	0.518

4. Selection

C4	1	0	0	1	0	0	0
C1	1	1	0	0	0	0	1

5. Crossover

				1					
C1	1	1	0	1	0	0	0	1	
C4	1	0	0	1	0	0	0	0	
				1					

⇒

1	1	0	1	0	0	0
1	0	0	0	0	0	1

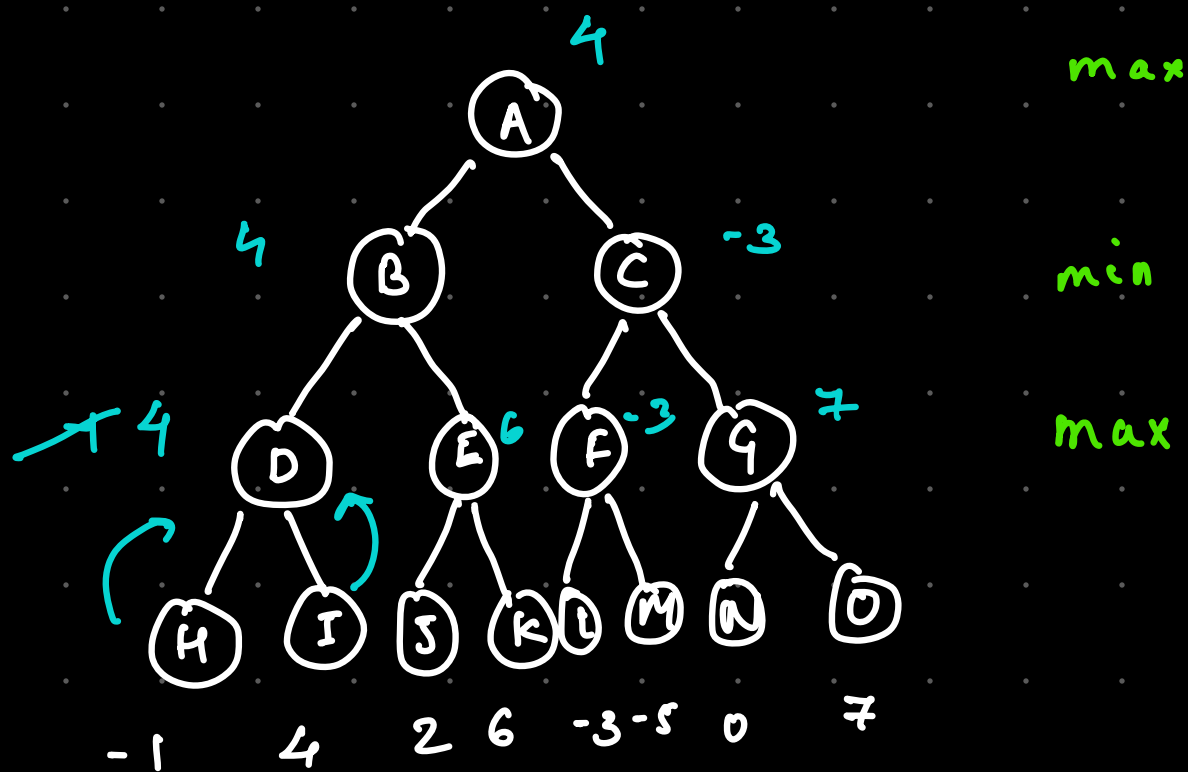
6. Mutation

								W _i x _i	P _i x _i
1	<u>1</u>	0	<u>0</u>	0	0	0		5	11
1	1	0	0	0	0	1		9	14

After first generation profit = 14

Adversarial = opp. Changing state at every step you don't want

Max $\Rightarrow$ DFS



$O(b^m)$ depth
branching factor

Alpha-Beta Pruning

↳ optimizer min-max

Ther. $\begin{cases} \alpha = -\infty & \text{MX at any pt. along max} \\ \beta = +\infty & \text{MN at } a \end{cases}$ ↗
↳

